ANDRII BOBROV

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RESEARCH INTERESTS

Monge–Kantorovich and Beckmann formulations of Optimal Transport.

ACADEMIC VISITS

The Fields Institute for Research in Mathematical Sciences Toronto, ON, Canada
Visiting Member (In-person) — Fully Funded Participant *Jul 2026 – Present*

- Selected for funded participation in the *Thematic Program on Optimal Transport* (travel & living support provided).
- Participating in advanced lecture series and participant-led working seminars on optimal transport.

EDUCATION

Kyiv Academic University (KAU) Kyiv, Ukraine
M.Sc. in Mathematics (Joint program with Institute of Mathematics, NAS of Ukraine) *Sep 2025 – Expected Jun 2027*

- **Advisor:** Georgii Riabov

National Technical University of Ukraine "KPI" Kyiv, Ukraine
B.Sc. in Applied Mathematics (Institute of Physics and Technology – IPT) *Sep 2021 – Jun 2025*

- **GPA:** 92/100.
- **B.Sc. Thesis:** *Optimal Transport on Graphs and the Beckmann Problem* (Advisor: Georgii Riabov).
- Established equivalence between W_1 optimal transport plans and Beckmann minimal flow problem on graphs; analyzed constructive flow-decomposition algorithms.

RESEARCH EXPERIENCE

Variational Formulations of the Monge–Kantorovich Problem in Banach Spaces
Master's Thesis Research (Joint work with G. Riabov) *Sep 2025 – Present*

- **Setup:** Investigating minimal L^1-norm vector fields satisfying $-\operatorname{div}(u\mu) = \rho\mu$ for probability measures satisfying a Poincaré inequality (in particular Gaussian measures), extending the Beckmann continuous transport formulation.
- **Finite-Dimensional Results:** Strong duality via Fenchel–Rockafellar duality; alignment $dU^* = \sigma\nabla w^*$. Continuity of σ is yet to be established (or disproved).

MANUSCRIPTS IN PREPARATION & WORK IN PROGRESS

- **A. Bobrov**, G. Riabov, *Beckmann problem on Wiener Space* (provisional title), manuscript in preparation.

TALKS & PRESENTATIONS

- **Kantorovich–Rubinstein Distance on Gaussian Spaces**, presented at:
  - *The Skorokhod Readings* (Online), Institute of Mathematics of NAS of Ukraine, Jun 2026.
  - *International Conference of Young Mathematicians* (Online), Institute of Mathematics of NAS of Ukraine, Jun 2026.
- **Barycenters in Wasserstein Space: Local Geometry and Parametric Convergence**, *KAU Mathematics Research Seminar*, Kyiv Academic University, Apr 2026.
- **On the Monge and Beckmann Formulations of Optimal Transport in Abstract Wiener Spaces**, *Optimal Transport Seminar*, The Fields Institute, Toronto, Canada (Oct 2026).

HONORS & ACADEMIC AWARDS

- **Fields Institute Visiting Support**, Thematic Program on Optimal Transport 2026
- **State Academic Excellence Scholarship**, KPI IPT (awarded to top 10% of students by GPA) 2021 – 2025

SELECTED ADVANCED COURSEWORK

Geometry & Topology: Lie Groups, Symmetries of Differential Equations, Algebraic Topology, Ergodic Theory, Topological Dynamics.

LANGUAGES

Ukrainian: Native | **English:** Professional working proficiency | **Japanese:** Intermediate / Conversational (∼N3) | **Russian:** Native